ASSIGNMENT-3 : CYBER SECURITY THREATS

GUIDED BY : VIPUL MISTRI

Q.1 List 5 different types of cyber security threats you might encounter while using popular apps like Instagram, WhatsApp, or Zomato, and give a real-life example for each.

ANS : **1. Phishing & Social Engineering (Smishing / Vishing)**

- **Concept:** Deceptive communications masquerading as trusted platforms or entities to trick users into revealing sensitive credentials, PINs, or One-Time Passwords (OTPs).

- **Real-Life Example (WhatsApp):** Attackers send a message on WhatsApp posing as official "WhatsApp Support" or a friend in distress, claiming they mistakenly sent a 6-digit verification code to your phone. If you share the code, the attacker registers your WhatsApp number on their device and locks you out of your account.

2. Malicious Ads & Malvertising

- **Concept:** Malicious advertisements or links embedded within social media platforms that direct users to compromised third-party pages or trick them into downloading malware.

- **Real-Life Example (Instagram):** Sponsored ads on Instagram promote "exclusive brand discounts" or "free gift cards." Clicking the ad redirects users to a lookalike external payment portal designed to harvest credit card details or download spyware onto mobile devices.

3. Account Takeover via Session Hijacking

- **Concept:** Intercepting active user session tokens or exploiting insecure authentication endpoints to gain unauthorized access to an existing logged-in account.

- **Real-Life Example (Zomato):** Connecting to unencrypted public Wi-Fi networks at a cafe allows an attacker using packet-sniffing tools on the same network to capture unencrypted session cookies or auth tokens, allowing them to

impersonate your session, view delivery addresses, and place orders using saved payment methods.

4. Man-in-the-Middle (MitM) Attacks

- **Concept:** A threat actor secretly intercepts and potentially alters communication between two parties (e.g., your mobile app and the backend server) who believe they are communicating directly.

- **Real-Life Example (WhatsApp / Messenger Apps):** Using rogue public Wi-Fi hotspots set up as "Free City Wi-Fi." If an app lacks strict SSL pinning, the rogue hotspot can intercept data in transit, exposing non-end-to-end encrypted app traffic, user metadata, or uploaded attachments.

5. Insecure Data Storage & Third-Party API Data Leaks

- **Concept:** Backend databases, third-party vendor APIs, or misconfigured cloud storage buckets exposing user personal data (PII) to unauthorized public scraping or indexing.

- **Real-Life Example (Zomato / Food Delivery Apps):** A misconfigured API endpoint or unsecured cloud storage database belonging to a third-party analytics or logistics partner leaks user phone numbers, names, and delivery addresses online, which scammers then buy to orchestrate targeted phishing calls.

Q.2 Simulate a phishing attack scenario by designing a fake login page for an app you use (for example, a mock Instagram or Flipkart login screen in HTML). Explain how a user could be tricked and what signs would help spot the threat. Hint: Do not collect or submit any real credentials—just create the page design and description.

ANS :

1. Mock Phishing Page Design (HTML)

In a real phishing campaign, attackers host a page that looks identical to a trusted brand's login portal. For this educational example, here is a harmless, static mock login screen resembling **Instagram**:

ASSIGNMENT-3 : CYBER SECURITY THREATS

GUIDED BY : VIPUL MISTRI

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Instagram • Login</title>
    <style>
        body {
            font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
            background-color: #fafafa;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .login-container {
            background: #ffffff;
            border: 1px solid #dbdbdb;
            padding: 40px;
            width: 350px;
            text-align: center;
            border-radius: 1px;
        }
```

```
.logo {
    font-family: 'Brush Script MT', cursive, sans-serif;
    font-size: 40px;
    margin-bottom: 30px;
    font-weight: bold;
}
.input-field {
    width: 100%;
    padding: 10px 8px;
    margin: 6px 0;
    background: #fafafa;
    border: 1px solid #dbdbdb;
    border-radius: 3px;
    box-sizing: border-box;
    font-size: 12px;
}
.login-btn {
    width: 100%;
    background-color: #0095f6;
    color: white;
    border: none;
    padding: 7px;
    border-radius: 4px;
    font-weight: 600;
    margin-top: 14px;
    cursor: pointer;
}
```

```
        .warning-banner {
            background: #fff3cd;
            color: #856404;
            padding: 10px;
            font-size: 11px;
            margin-bottom: 15px;
            border: 1px solid #ffeeba;
            border-radius: 3px;
        }
    </style>
</head>
<body>

    <div class="login-container">
        <!-- Educational Warning Banner -->
        <div class="warning-banner">
            <strong>Simulation Only:</strong> This is a mock educational interface. Do not enter real credentials.
        </div>

        <div class="logo">Instagram</div>

        <!-- Form points nowhere (action="#") to ensure safety -->
        <form action="#" method="POST">
            <input type="text" class="input-field" name="username" placeholder="Phone number, username, or email" required>
            <input type="password" class="input-field" name="password" placeholder="Password" required>
```

```
        <button type="submit" class="login-btn">Log In</button>
      </form>
    </div>


</body>
</html>
```

2. How the User Gets Tricked

1. **The Lure:** The victim receives a direct message or email stating: *"Your account has been flagged for copyright review. Verify your identity within 24 hours to prevent permanent deletion."*
2. **The Link:** The message contains a link disguised to look official (e.g., help-instagram-support-center.com).
3. **The Trap:** Clicking the link loads the mock page shown above. Because the logos, input boxes, and styling mimic the authentic brand, the user feels comfortable and enters their username, password, and two-factor code, believing they are securing their account.

3. Key Red Flags to Spot the Threat

- **Suspicious URL / Domain Name:** The browser's address bar displays an odd domain name (e.g., instagram.support-login-auth.net) instead of the legitimate root domain (instagram.com).
- **Lack of Contextual Navigation:** On phishing pages, secondary links (like "Forgot password?", "Sign up", or footer privacy policy links) are often broken, dead buttons, or redirect back to the phishing form rather than actual help pages.
- **Urgency and Coercion:** Phishing lures almost always rely on artificial deadlines (e.g., "within 24 hours" or "immediate suspension") to trigger panic and bypass critical thinking.
- **Missing Official App Integration:** Official login flows on mobile devices often prompt biometric checks or use native operating system authentication pop-ups rather than basic web forms.

Q.3 Research and write a short report (150-200 words) on a recent cyber attack that affected a major Indian tech platform (such as Paytm, IRCTC, or PhonePe). Describe the type of threat, how it was carried out, and what was done to fix it.

ASSIGNMENT-3 : CYBER SECURITY THREATS

GUIDED BY : VIPUL MISTRI

ANS : **Incident Report: Star Health Insurance Data Exfiltration (2025)**

Overview & Threat Type: In May 2025, Star Health Insurance—one of India's largest health insurance providers—experienced a massive unauthorized API data exfiltration and cyber-extortion incident. The attack compromised sensitive personal identifiable information (PII), policy details, and medical records of millions of customers.

How It Was Carried Out: Threat actors identified and exploited misconfigured, exposed backend database API endpoints. This allowed them to bypass standard authentication controls and silently query and extract massive volumes of customer records from the backend infrastructure. To monetize and distribute the stolen data, the attackers created automated, custom Telegram bots that allowed public users to search and download leaked insurance documents on demand.

Remediation & Fixes: Upon detecting the exfiltration, Star Health and cybersecurity response teams implemented immediate countermeasures:

- **Endpoint Isolation:** Unsecured API endpoints were immediately taken offline and patched with strict authentication and access-control policies.
- **Legal & Regulatory Action:** The organization filed formal complaints with law enforcement agencies and CERT-In, securing court orders to force Telegram to take down the automated data-sharing channels.
- **Infrastructure Auditing:** The platform conducted comprehensive third-party security audits, implemented strict multi-factor authentication (MFA) across internal controls, and introduced continuous API threat-monitoring systems.

Q.4 Pick any one cyber security threat (like ransomware, phishing, or DDoS) and create a simple infographic or poster (hand-drawn or digital) that explains how it works, who is at risk, and how to prevent it.

ASSIGNMENT-3 : CYBER SECURITY THREATS

GUIDED BY : VIPUL MISTRI

ANS :



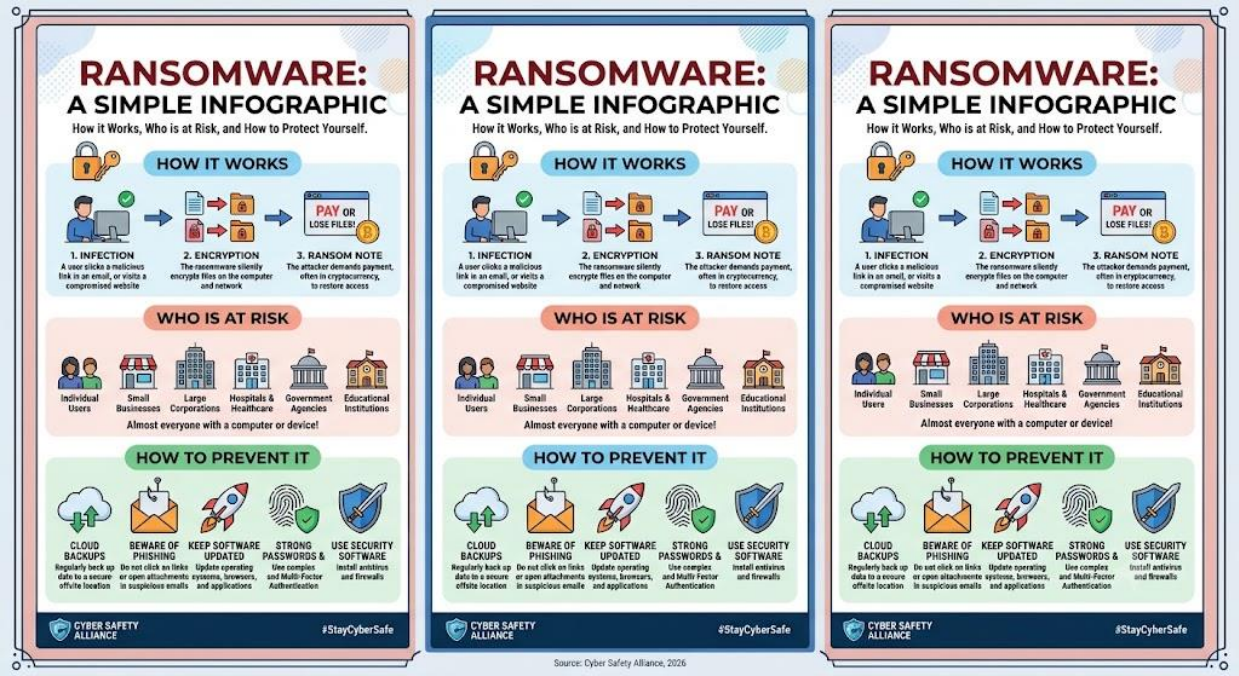
Here is a simple infographic illustrating the lifecycle of a Ransomware attack, identifying potential victims, and detailing essential prevention strategies. Ransomware is one of the most disruptive cybersecurity threats facing individuals and organizations today, combining encryption technology with extortion tactics.

The design breaks down the threat into three actionable sections:

- **Lifecycle:** Shows how an initial infection leads to silent encryption and culminates in a ransom demand.
- **Potential Targets:** Emphasizes that from individual users to critical infrastructure, almost anyone with a computer or device is at risk.
- **Prevention Checklist:** Highlights key defensive habits, such as regular backups, phishing awareness, software updates, strong authentication, and security software.

ASSIGNMENT-3 : CYBER SECURITY THREATS

GUIDED BY : VIPUL MISTRI



Based on our recent research into ransomware, I have generated a simple infographic poster summarizing the critical aspects of this threat.

The infographic is structured into three clear sections:

1. **How It Works:** A visual walkthrough of the attack chain, from initial infection (e.g., clicking a malicious link) to the encryption of files and the delivery of the ransom note.
2. **Who Is At Risk:** Highlights that almost everyone is a potential target, from individual users to large corporations, hospitals, government agencies, and educational institutions.
3. **How to Prevent It:** A checklist of essential preventative measures, including maintaining offline backups, strong phishing awareness, timely software updates, robust authentication, and the use of dedicated security software.